

Journal Pre-proof

Principal Results of the VITamin D and OmegA-3 TriaL (VITAL) and Updated Meta-analyses of Relevant Vitamin D Trials

JoAnn E. Manson, Shari S. Bassuk, Julie E. Buring, for the VITAL Research Group



PII: S0960-0760(19)30542-4

DOI: <https://doi.org/10.1016/j.jsbmb.2019.105522>

Reference: SBMB 105522

To appear in: *Journal of Steroid Biochemistry and Molecular Biology*

Received Date: 12 September 2019

Revised Date: 22 October 2019

Accepted Date: 29 October 2019

Please cite this article as: Manson JE, Bassuk SS, Buring JE, Principal Results of the VITamin D and OmegA-3 TriaL (VITAL) and Updated Meta-analyses of Relevant Vitamin D Trials, *Journal of Steroid Biochemistry and Molecular Biology* (2019), doi: <https://doi.org/10.1016/j.jsbmb.2019.105522>

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**Principal Results of the VITamin D and OmegA-3 Trial (VITAL)
and Updated Meta-analyses of Relevant Vitamin D Trials**

JoAnn E. Manson, MD, DrPH; Shari S. Bassuk, ScD; Julie E. Buring, ScD; for the VITAL
Research Group*

Author affiliations:

Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA
(JEM, SSB, JEB), Department of Epidemiology (JEM, JEB), Harvard T.H. Chan School of Public
Health, Boston, MA

*A complete list of the members of the VITAL Research Group is provided in the Appendix.

Corresponding Author: JoAnn E. Manson, MD, DrPH, Department of Medicine, Brigham and
Women's Hospital, Harvard Medical School, 900 Commonwealth Avenue, 3rd Fl., Boston, MA;
telephone: 617-278-0871; fax: 617-731-3843; e-mail: jmanson@rics.bwh.harvard.edu

Prepared for: Journal of Steroid Biochemistry and Molecular Biology

Word count, abstract: 201

Word count, main text: 3,523

Tables and Figures: 7

HIGHLIGHTS

- In VITAL, vitamin D did not reduce the primary cancer or cardiovascular endpoints [83
characters]

- Vitamin D reduced total cancer mortality in analyses excluding early follow up [80 characters]
- Meta-analyses of trials show a reduction in cancer mortality but not incidence [80 characters]
- More research is needed to identify those most likely to benefit from vitamin D [81 characters]

ABSTRACT

Whether supplemental vitamin D reduces risk of cancer or cardiovascular disease (CVD) is relatively unexplored in randomized trial settings. The *VITamin D and Omega-3 Trial* (VITAL) was a nationwide, randomized, placebo-controlled, 2x2 factorial trial of daily vitamin D₃ (2000 IU) and marine omega-3 fatty acids (1 g) in the primary prevention of cancer and CVD among 25,871 U.S. men aged ≥50 and women aged ≥55, including 5,106 African Americans. Median treatment duration was 5.3 years. Vitamin D did not significantly reduce the primary endpoint of total invasive cancer incidence (hazard ratio [HR]=0.96 [95% confidence interval 0.88-1.06]) but showed a promising signal for reduction in total cancer mortality (HR=0.83 [0.67-1.02]), especially in analyses that accounted for latency by excluding the first year (HR=0.79 [0.63-0.99]) or first 2 years (HR=0.75 [0.59-0.96]) of follow-up. Vitamin D did not significantly reduce the co-primary endpoint of major CVD events (HR=0.97 [0.85-1.12]), other cardiovascular endpoints, or all-cause mortality (HR=0.99 [0.87-1.12]). Updated meta-analyses that include VITAL and other recent vitamin D trials indicate a significant reduction in cancer mortality but not in cancer incidence or CVD endpoints. Additional research is needed to determine which

individuals may be most likely to derive a net benefit from vitamin D supplementation. (VITAL clinicaltrials.gov identifier: NCT01169259)

Key words: cancer; cardiovascular disease; primary prevention; race and ethnicity; randomized controlled trial; vitamin D.

1. INTRODUCTION

The VITamin D and Omega-3 Trial (VITAL) is the only completed large ($N \geq 10,000$) trial of moderate- to high-dose vitamin D for cancer and cardiovascular disease (CVD) prevention to date. We summarize the trial's rationale, design, and primary results, which have been previously reported and discussed in greater detail [1-3], as well as findings from recent meta-analyses of randomized trials of vitamin D that included VITAL data [4, 5].

2. VITAL OVERVIEW

2.1. Rationale. An established intervention for the prevention and treatment of bone disorders [6], supplemental vitamin D has more recently also been used for the possible prevention of cancer and CVD. In the U.S., vitamin D sales began climbing more than a decade ago [7, 8], and vitamin D now ranks among the most widely used supplements [9, 10]. These potential benefits were supported by promising data from some ecologic, laboratory, and observational studies, but such data were inconsistent and insufficient to establish causality [1, 6, 11]. Vitamin D trials that had examined cancer or CVD outcomes in secondary or *post hoc* analyses had yielded largely null results but low doses, inadequate power, short durations, and/or suboptimal assessment of endpoints were limitations [6]. Large trials of doses high enough to yield meaningful changes in 25-hydroxyvitamin D (25(OH)D) levels and with cancer or CVD as primary endpoints did not exist. Thus, the Institute of Medicine in 2011 [6] and the U.S. Preventive Services Task Force in 2014 [12] concluded that available data did not allow a

definitive assessment of effectiveness and benefit-risk balance of vitamin D supplementation for these purposes. The Institute of Medicine called for vitamin D trials at high doses (at least twice the current recommended dietary allowance [RDA] of 600-800 IU/d for bone health) in diverse populations, including African Americans, who tend to have lower cutaneous synthesis of vitamin D in response to solar radiation than members of other racial groups [13]. We proposed and carried out VITAL to fill this knowledge gap.

2.2. Design. VITAL was a nationwide, randomized, double-blind, placebo-controlled trial of the benefits and risks of supplemental vitamin D₃ (2000 IU/d) and marine omega-3 fatty acids (1 g/d Omacor® fish-oil capsule with 840 mg of omega-3 fatty acids, including eicosapentaenoic acid [EPA, 460 mg] + docosahexaenoic acid [DHA, 380 mg]) in the primary prevention of cancer and CVD among 25,871 U.S. men and women, aged ≥50 and ≥55, respectively [1, 2, 14, 15]. By design, a similar number of men and women were enrolled, and African Americans were oversampled. Eligible participants had no history of cancer (except non-melanoma skin cancer), myocardial infarction, stroke, transient ischemic attack, or coronary revascularization. They were required to agree to limit daily intakes of vitamin D and calcium from all supplemental sources, including multivitamins, to no more than 800 IU and 1200 mg, respectively (the recommended dietary allowances [RDAs] for older adults [6]) and to avoid the use of fish-oil supplements. Individuals with renal failure or dialysis, severe liver disease, anticoagulant use, history of hypercalcemia or parathyroid disorders, or other conditions that could pose safety concerns were not allowed to participate. After successfully completing a 3-month placebo run-in, participants were randomized to vitamin D, omega-3 fatty acids, both active agents, or both placebos in a 2x2 factorial design (Figure 1). Randomization took place from November 2011 to March 2014. Randomized treatment ended as scheduled on 31 December 2017. The median intervention period was 5.3 years, with a range of 3.8 to 6.1 years.

Participants completed baseline questionnaires regarding clinical and lifestyle risk factors for cancer, CVD, and other conditions. They also completed annual follow-up questionnaires

regarding treatment compliance and side effects, risk factor updates, and endpoint occurrence. Study physicians blinded to treatment assignment reviewed participants' medical records to confirm or disconfirm reported endpoints using accepted criteria [16-18]. Deaths were ascertained via the National Death Index-Plus and other sources. All participants were asked to provide an optional pre-randomization blood sample. A total of 16,956 participants (66% of the study population) did so; ~6000 of these participants also provided follow-up samples in years 1-5. A sample of Boston-area participants (n=1,054) visited a local Clinical and Translational Science Center clinic for detailed assessments at baseline and years 1, 2, and 4. With the exception of the clinic assessments, blood collections, and certain ancillary studies, VITAL has been carried out primarily by postal and electronic communication.

2.3. Baseline characteristics. Characteristics of the study population are shown in Table 1. Of the 25,871 participants, 49% are men and 51% women, with a mean age of 67.1 years. The cohort is racially diverse, with 71% self-declared non-Hispanic whites, 20% African Americans, and the rest members of other racial/ethnic groups. As expected in this large sample, randomization balanced the characteristics between the treatment groups, minimizing the likelihood that results are attributable to known or unknown confounding factors [14].

2.4. Follow-up, adherence, achieved 25-hydroxyvitamin D levels. During the 5.3-year trial, yearly questionnaire response rates averaged 93%, and mortality follow-up exceeded 98% [2]. The percentage of participants who reported taking at least two-thirds of their study capsules averaged 80% during the intervention period; and fewer than 5% and 10% of participants reported outside vitamin D use at 2 and 5 years, respectively [2]. Biomarker analyses support high adherence to study medication guidelines, showing a large post-randomization difference in total serum 25-hydroxyvitamin D (25(OH)D) levels between the vitamin D and placebo groups throughout the trial. Among the ~15,500 participants with analyzable baseline blood samples, the mean ($\pm$ SD) 25(OH)D level at trial entry was 30.9 $\pm$ 10.0 ng/mL (77 $\pm$ 25 nmol/L) in the active vitamin D group and 30.8 $\pm$ 10.0 ng/mL (77 $\pm$ 25 nmol/L) in the placebo group. Of these

participants, 13% had 25(OH)D levels <20 ng/mL (<50 nmol/L) and 32% had levels of 20-<30 ng/mL (50-<75 nmol/L). In the subgroup with at least one additional measure at years 1, 2, 4, or 5, the achieved 25(OH)D at each follow-up time point was ≥ 41 ng/mL (102.5 nmol/L) in the active vitamin D group and 29-31 ng/mL (72.5-77.5 nmol/L) in the placebo group.

3. VITAMIN D AND CANCER

3.1. VITAL results. In VITAL, vitamin D did not significantly reduce the primary endpoint of total invasive cancer incidence (hazard ratio [HR]=0.96 [95% confidence interval 0.88-1.06]) but showed a promising signal for reduction in total cancer mortality (HR=0.83 [0.67-1.02]), especially in analyses that accounted for latency by excluding the first year (HR=0.79 [0.63-0.99]) or first 2 years (HR=0.75 [0.59-0.96]) of follow-up (Table 2) [2]. The cumulative incidence curves for cancer mortality began to diverge clearly at 4 years (Figure 2). The HRs for incidence of prespecified site-specific cancers were 1.02 (0.79-1.31) for breast cancer, 0.88 (0.72-1.07) for prostate cancer, and 1.09 (0.73-1.62) for colorectal cancer.

Although vitamin D did not reduce total cancer incidence overall, prespecified analyses of subgroups produced intriguing findings. The following characteristics, assessed at baseline, were examined as potential effect modifiers: age; sex; race/ethnicity; traditional cardiovascular risk factors (smoking, diabetes, hypertension, high cholesterol, and parental history of premature MI); body mass index (BMI); aspirin use; statin use; serum 25(OH)D; outside use of supplemental vitamin D; and concurrent randomization to the active omega-3 fatty acid group. Selected findings are shown in Table 3. Individuals with normal BMI (<25 kg/m²) experienced a significant treatment-associated reduction in cancer risk (HR=0.76 [0.63-0.90]), but overweight or obese individuals did not (p, interaction=0.002). African Americans assigned to vitamin D also had a suggestive reduction in cancer risk (HR=0.77 [0.59-1.01]), although the p-value for interaction by race/ethnicity was not significant (p, interaction=0.21). The association between vitamin D and cancer incidence did not significantly vary by any of the other examined

characteristics; nor did vitamin D significantly reduce cancer risk in other examined subgroups. In preliminary analyses of cancer stage at diagnosis, the incidence of metastatic cancer, advanced cancer, or both was slightly lower in the vitamin D group than in the placebo group, but differences were not significant.

3.2. Discussion. VITAL is the first large ($N \geq 10,000$) trial of moderate- or high-dose vitamin D for cancer prevention. Despite methodologic limitations, several earlier trials had also suggested stronger benefits for cancer mortality than for cancer incidence. Among the largest of these trials are the Women's Health Initiative (WHI) calcium-vitamin D trial [19-21], the Randomized Evaluation of Calcium or vitamin D (RECORD) trial [22], and a UK trial by Trivedi et al. [23]. The WHI randomized >36,000 US postmenopausal women to 7 years of daily calcium (1000 mg) plus vitamin D₃ (400 IU) or to placebo and found a suggestion of a protective effect against cancer mortality (HR=0.89 [0.77-1.03]) but not cancer incidence (HR=0.98 [0.91-1.05]) [22]; the below-RDA dose is a limitation. In the RECORD trial, 5,292 UK adults aged ≥ 70 were randomized to daily vitamin D₃ (800 IU), calcium (1000 mg), both, or placebo for 2 to 5.2 years for secondary fracture prevention and then followed observationally for 3 years; vitamin D appeared to reduce cancer mortality (HR=0.85 [0.68-1.06]) but not cancer incidence (HR=1.07 [0.92-1.25]) [22]. Trivedi et al. [23] randomized 2,686 older adults to vitamin D₃ (100,000 IU every 4 months [~ 833 IU/d]) or placebo for up to 5 years and also found a nonsignificant inverse association for cancer mortality (HR=0.86 [0.61-1.20]) but not cancer incidence (HR=1.09 [0.86-1.36]); the trial's modest size and intermittent bolus dosing, which has been associated with nonphysiological fluctuations in vitamin D blood levels [24], are limitations. On the other hand, the 3.3-year Vitamin D Assessment Study (ViDA), which tested high-dose vitamin D (100,000 IU/month [~ 3300 /d]) vs. placebo for CVD prevention in 5,110 New Zealanders, found no signal of benefit for either cancer mortality (HR=0.99 [0.60-1.64]) or cancer incidence (HR=1.01 [0.81-1.25]) [25], although the short duration and intermittent bolus dosing limit firm conclusions. In a 2019 meta-analysis of randomized trials that included VITAL and these and other earlier trials,

treatment-associated RRs for cancer mortality (5 trials, 6547 cancer deaths) and cancer incidence (10 trials, 6,547 incident cancers) were 0.87 (0.79-0.96) and 0.98 (0.93-1.03), respectively (Figure 3) [4].

Plausible mechanisms by which vitamin D may reduce cancer incidence and/or cancer mortality are shown in Figure 4. Laboratory research suggesting that $1,25(\text{OH})_2\text{D}$, the active vitamin D hormone produced from $25(\text{OH})\text{D}$, decreases tumor invasiveness, angiogenesis, and metastatic propensity [26, 27]. The cancer mortality findings are also supported by observational studies showing that higher $25(\text{OH})\text{D}$ levels at cancer diagnosis predict longer survival in patients [28-34]. In some [35-45], but not all [46-50], studies, higher pre-diagnostic $25(\text{OH})\text{D}$ or genetically determined vitamin D levels in initially cancer-free cohorts favorably correlated with cancer mortality and/or survival following diagnosis. Two recent trials in patients with gastrointestinal cancer suggested that high-dose vitamin D may reduce disease progression and death [51, 52]. However, not all vitamin D trials in cancer patients have shown such benefits [53, 54], and initiation of treatment in the preclinical stage or earlier (as in VITAL) may have advantages. Subgroup considerations are discussed in Section 5.

4. VITAMIN D AND CARDIOVASCULAR DISEASE

4.1. VITAL results. Vitamin D did not reduce the co-primary endpoint of major CVD events (a composite of MI, stroke, and CVD mortality; $\text{HR}=0.97$ [0.85-1.12]) [2]. It also did not reduce prespecified secondary cardiovascular endpoints, including an expanded composite of major CVD events plus coronary revascularization ($\text{HR}=0.96$ [0.86-1.08]), or myocardial infarction ($\text{HR}=0.96$ [0.78-1.19]), stroke ($\text{HR}=0.95$ [0.76-1.20]), and CVD mortality ($\text{HR}=1.11$ [0.88-1.40]) considered individually. Vitamin D had no effect on all-cause mortality ($\text{HR}=0.99$ [0.87-1.12]). Similar findings were seen in analyses that excluded the first year or two years of follow-up or that censored for noncompliance. There were no significant treatment-associated increases in risk of hypercalcemia, kidney stones, or gastrointestinal symptoms. Vitamin D did not influence 1-year changes in lipids or inflammatory markers [55].

The association between vitamin D and risk of CVD endpoints or all-cause mortality did not differ significantly by race/ethnicity, cardiovascular risk factors, serum 25(OH)D level, concurrent randomization to omega-3 fatty acids, or other characteristics prespecified as potential effect modifiers (see Section 3.1); and vitamin D did not significantly reduce these endpoints in any subgroup. Selected findings are shown in Table 3.

4.2. Discussion. VITAL is the first large trial of moderate- or high-dose vitamin D for CVD prevention. Its null cardiovascular findings agree with the results of earlier trials. In the WHI, which tested daily calcium (1000 mg) plus low-dose vitamin D₃ (400 IU), HRs for coronary heart disease, stroke, and CVD mortality were 1.04 (0.92-1.18), 0.95 (0.82-1.10), and 0.92 (0.77-1.10), respectively [20, 56]. In RECORD, which tested daily vitamin D (800 IU/d), HRs for myocardial infarction, stroke, and vascular disease mortality were 0.97 (0.75-1.26), 1.06 (0.85-1.32), and 0.91 (0.79-1.05), respectively [22, 57]. The UK trial of Trivedi et al. [23] reported HRs of 0.94 (0.77-1.15), 0.84 (0.56-1.27), 0.90 (0.77-1.06), and 0.84 (0.65-1.10) for coronary heart disease incidence, coronary heart disease mortality, CVD incidence, and CVD mortality, respectively, with vitamin D (100,000 IU every 4 months). Not surprisingly, meta-analyses of these and smaller vitamin D trials, even those that considered only trials of RDA-level or higher doses (≥ 800 IU/d) [58], have not shown significant cardiovascular benefit [57-61]. More recently, ViDA also reported a lack of CVD benefit for high-dose vitamin D (MI: RR=0.90 [0.54-1.50]; stroke: RR=0.95 [0.55-1.62]) [62], although, as noted earlier, the trial's short duration and bolus dosing (100,000 IU/month) are limitations. In a 2019 meta-analysis of vitamin D trials that included VITAL and ViDA, vitamin D did not reduce risk of major adverse cardiovascular events (10 trials, 6243 events, 79,111 participants; RR=1.00 [0.95-1.06]), myocardial infarction (18 trials, 2550 events, 82,576 participants; RR=1.00 [0.93-1.08]), stroke (15 trials, 2354 events, 82,239 participants; RR=1.06 [0.98-1.15]), or cardiovascular mortality (10 trials, 2202 events, 76,783 participants; RR=0.98 [0.90-1.07]) [5]. There was little evidence of effect modification by

baseline 25(OH)D level, vitamin D dose or administration frequency, or concurrent randomization to supplemental calcium.

In agreement with VITAL, ViDA found that high-dose vitamin D did not reduce the risk of all-cause mortality [62]. Trials of lower doses also show neutral effects or at most modest reductions in this endpoint [59-61]. In a 2019 meta-analysis of 20 vitamin D trials (6,502 deaths among 83,059 participants) that included VITAL and ViDA, the treatment-associated RR for all-cause mortality was 0.97 (0.93-1.02) [5]. Longer follow-up may necessary to observe benefit, however.

The neutral cardiovascular findings among normal-weight and African-American participants contrast with the significant and suggestive findings, respectively, for cancer reduction in these groups. This divergence may be attributable to differing vitamin D requirements for CVD and cancer prevention. In observational studies, individuals with 25(OH)D levels of 20-25 ng/mL tend to have the lowest CVD risk [63], whereas those with levels above 30 ng/mL tend to have the lowest cancer (at least colorectal cancer) risk [64]. Thus, many participants in VITAL (and other trials) may have enrolled with their vitamin D requirements for cardiovascular health already fulfilled. Although neither VITAL nor ViDA found that vitamin D reduced CVD risk among participants with low baseline 25(OH)D, a trial restricted to individuals with 25(OH)D levels well below the 20 ng/mL recommended for bone health [6] might demonstrate greater benefit. However, targeting those with known vitamin D deficiency for participation in a long-term trial with a 50% chance of randomization to placebo would not be ethical.

Mechanisms by which vitamin D may prevent CVD are shown in Figure 4. Results of laboratory and animal studies suggest that $1,25(\text{OH})_2\text{D}$ inhibits vascular smooth muscle cell proliferation and vascular calcification, favorably affects volume homeostasis and blood pressure via regulation of the renin-angiotensin-aldosterone system, reduces inflammation, and improves insulin sensitivity [1, 65-67]. In prospective observational studies, 25(OH)D levels are

inversely correlated with cardiovascular risk factors as well as CVD events [63, 68, 69]. On the other hand, small, short-term trials of vitamin D have generally failed to show convincing improvement in intermediate cardiovascular endpoints, including blood pressure, glucose or insulin homeostasis, inflammation, vascular function parameters, and lipids [3]. Indeed, in VITAL, as noted earlier, vitamin D did not improve lipids or inflammatory markers after one year. Analyses of intervention effects on other CVD-related endpoints are underway in VITAL (Section 6).

5. SUBGROUP CONSIDERATIONS

As noted earlier, supplemental vitamin D appeared to reduce risk of cancer (though not CVD) in individuals with normal BMI and in African Americans in VITAL. On the other hand, the effect of supplementation on risk of cancer and CVD did not vary according to baseline levels of total 25(OH)D. These findings are worthy of further study.

5.1. Novel vitamin D biomarkers. Total 25(OH)D has long been considered to be the single best indicator of clinical vitamin D status. However, nearly all circulating 25(OH)D is bound to either to vitamin D binding protein (DBP, ~85%) or albumin (~15%) [70]. Bioavailable 25(OH)D is defined as 25(OH)D not bound to DBP, and free 25(OH) is defined as 25(OH)D not bound to DBP or albumin. Some studies (though not all [70, 71]) suggest that DBP, bioavailable 25(OH)D, free 25(OH)D, and parathyroid hormone may be more biologically relevant markers than—or useful complements to—total 25(OH)D in assessing vitamin D status [70, 72] as a predictor of clinical outcomes such as cancer [73-75] and coronary heart disease [76-78]. Whether baseline levels of (or changes in) these markers affect the association between vitamin D supplementation and cancer or CVD warrants additional investigation and is currently under study in VITAL.

5.2. Obesity. Several lines of evidence suggest that vitamin D bioactivity is altered in obesity. Excess body weight is associated with lower levels of free and total 25(OH)D [79]; may disproportionately lower the former [80]; and may influence the correlation between the two

markers [81]. In rodents, diet-induced obesity decreased free but not total 25(OH)D, and increased expression of *CYP2R1*, a key vitamin D-related gene [80]. Higher free 25(OH)D levels predicted cardiovascular and all-cause mortality in coronary heart disease patients with normal BMI but not in patients who were overweight or obese [78]. Randomized trials have found more favorable effects of vitamin D supplementation on blood pressure [82] and incident diabetes [83] in normal-weight individuals than in their heavier counterparts. Taken together, these data support the validity of VITAL's finding of an interaction between treatment and BMI on incident cancer.

5.3. Genetic factors. Clarifying the role of genetic factors may be helpful in understanding VITAL's promising signal of a treatment-associated cancer reduction in African Americans. For example, people with African ancestry differ from those with European ancestry with respect to variation in *GC* [84, 85], a gene that encodes DBP, which, as noted earlier, is the major transporter of circulating vitamin D. Genetic variation in *GC*, *VDR*, and other genes related to vitamin D synthesis, transport, and metabolism [86, 87] may interact with supplemental vitamin D to affect risk of cancer, CVD, and other outcomes. These potential interactions have rarely been examined in a randomized trial setting. However, results from the Vitamin D/Calcium Polyp Prevention Study, which randomized 2,259 participants to vitamin D, calcium, both, or placebo for prevention of recurrent colorectal adenoma, point to the potential utility of such analyses [88]. In that trial, although vitamin D had no effect on the risk of advanced colorectal adenoma in the total cohort, analyses stratified by *VDR* variants revealed a 64% treatment-associated reduction in risk among participants with the rs7968585 AA genotype and a 41% increase in risk among those with 1 or 2 G alleles.

6. BENEFIT-RISK BALANCE, POST-INTERVENTION FOLLOW-UP

In VITAL, results of ancillary investigations of colorectal adenoma, mammographic density, telomere biology, diabetes, hypertension, heart failure, atrial fibrillation, and cardiac structure/function (2D-echocardiograms) will soon provide a fuller picture of cancer- and

cardiovascular-related effects of daily high-dose vitamin D supplementation. In addition, forthcoming results of ancillary studies of other outcomes, including fractures, falls, cognition, depression, infections, autoimmune disorders, respiratory conditions, and eye disease, will contribute to an assessment of the overall balance of benefits and risks of such supplementation. Ongoing post-intervention follow-up of the cohort will permit an assessment of potential latent and long-term treatment effects and will increase statistical power, an especially important consideration for analyses of secondary endpoints and subgroup effects.

7. OTHER LARGE VITAMIN D TRIALS

VITAL tested only one vitamin D dose and administration frequency and thus could not compare the efficacy of differing doses or of daily vs. less frequent dosing. Ongoing vitamin D trials [89] may help resolve these issues. However, to our knowledge, there is only one planned, ongoing, or completed large ($N \geq 10,000$) trial of moderate- or high-dose vitamin D other than VITAL. In D-Health [90], 21,315 Australians aged 65-84 have been randomized to 5 years of vitamin D (60,000 IU/month) or placebo and are being followed for a cumulative total of 10 years (via national registry linkage) for incident cancer and all-cause mortality; CVD will also be examined. Trial-phase results are expected in 2021. Of note, the D-Health study population includes few if any black participants, precluding an examination of supplementation effects in this group.

8. CONCLUSION

In VITAL, daily high-dose vitamin D supplementation did not significantly reduce the trial's co-primary endpoints of total invasive cancer and major CVD events. However, vitamin D significantly reduced total cancer mortality in analyses excluding early follow-up. Meta-analyses of randomized trials that include VITAL data show similar results. Additional investigation is necessary to determine which individuals may be most likely to derive a net benefit from supplementation.

ACKNOWLEDGMENTS

VITAL Investigators, Staff, and Study Participants

The authors thank the VITAL investigators, staff, and study participants for their dedication and commitment to the trial.

Funding/Support: VITAL is an investigator-initiated trial supported by grants U01 CA138962 and R01 CA138962, which include support from the National Cancer Institute, National Heart, Lung and Blood Institute, Office of Dietary Supplements, National Institute of Neurological Disorders and Stroke, and the National Center for Complementary and Integrative Health. The ancillary studies are supported by grants from multiple Institutes, including the National Heart, Lung and Blood Institute; the National Institute of Diabetes and Digestive and Kidney Diseases; the National Institute on Aging; the National Institute of Arthritis and Musculoskeletal and Skin Diseases; the National Institute of Mental Health; and others.

Pharmavite LLC of Northridge, California (vitamin D) and Pronova BioPharma of Norway and BASF (Omacor fish oil) donated the study agents, matching placebos, and packaging in the form of calendar packs. Quest Diagnostics (San Juan Capistrano, CA) measured the serum 25(OH)D levels and plasma omega-3 index at no cost to the study.

VITAL has been approved by the Institutional Review Board of Partners Healthcare/Brigham and Women's Hospital. The study agents have received Investigational New Drug Approval from the U.S. Food and Drug Administration.

VITAL is registered at [clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01169259) (NCT01169259). The VITAL website is www.vitalstudy.org.

Role of the Sponsor: The NIH sponsors of VITAL had a role in the design and conduct of the study and interpretation of the data. Final decisions concerning the above, however, as well as data collection, management, analysis, manuscript review or approval, and decision to submit the manuscript for publication resided with VITAL investigators and the VITAL research group. The opinions expressed in the manuscript are those of the study authors and do not necessarily represent the views of the Department of Health and Human Services/National Institutes of Health.

Conflict of Interest Disclosures: All authors will submit a completed ICMJE Form for Disclosure of Potential Conflicts of Interest.

Appendix

VITAL Research Group:

VITAL Steering Committee: JoAnn E. Manson (Chair), Julie E. Buring (Chair), Nancy R. Cook, I-Min Lee, William Christen, Shari S. Bassuk, Samia Mora, Heike Gibson, David Gordon, Trisha Copeland, Denise D'Agostino, Georgina Friedenber, Claire Ridge, Vadim Bubes, Edward L. Giovannucci, Walter C. Willett (all at Brigham and Women's Hospital, Harvard Medical School, Boston; Drs Manson, Buring, Cook, Lee, Giovannucci and Willett are also at the Harvard T.H. Chan School of Public Health).

Scientific consultants: John Baron (University of North Carolina, Chapel Hill), Michael Holick (Boston Medical Center), Bruce Hollis (University of South Carolina).

Other Members of the VITAL Research Group: (Brigham and Women's Hospital): Christine M. Albert, Diane Gold, Meryl LeBoff, Olivia Okereke, Aruna Pradhan, Howard Sesso, Wendy Chen, Paulette Chandler, J. Michael Gaziano; Olga Demler, Kathryn Rexrode, Karen Costenbader, John Forman, Erik Alexander, Sonia Friedman, Jeffrey Katz, Shumin Zhang, Jennifer Lin, Joseph Walter, Julie Duszlak, Kate Kalan, Jean MacFadyen, Natalya Gomelskaya, David Bates, Ara Sarkissian, Mary Breen, Yeulolani Andrade, Manickavasagar Vinayagamoorthy, Chunying Li, Eunjung Kim, Franco Giulianini, Gregory Kotler, Marty Van Denburgh, Rimma Dushkes, Yanyan Liu, Eduardo Pereira, Lisa Fields-Johnson, George Menjin, Lucy Liu, Lauren Girard, Scott Zeller, Naomi Riches, Katelyn Hasson, Ellen Bhang, Maria Revilla, Elena McCarthy, Alex Moran, Kristen Haise, Leah Arsenault, Philomena Quinn, Sancia Grimes, Ivan Fitchorov, Kurt Schwerin, Shamikhah Curry, Annie Murray, Angela Zhang, Diana Walrond-Williams, Alison Weinberg, Chris Pfeffer, Margarette Haubourg, Viviane Nguyen, Henry Ouellette, Rolando Rodriguez, Tony Montgomery, Keith Morse, Vincent Guzman, Megan Perry, Sandra Weekes, Doug Smith, Allison Clar, Sara Curran, Yaneve Fonge, David Hibbert, Louisa Paine, Kelly Royce, Courtney Splaine, Jennifer McMahon, David Eldridge, Laura Hand, Kay Inandan, Meghan Rieu Werden, Harriet Samuelson, Andrea Hrbek, Megan Mele, Eileen Bowes, Mary Anne Ryan

(Massachusetts General Hospital, Boston): Carlos Camargo, Jacqueline Danik, Ravi Thadhani

(Vanderbilt University, Nashville): Thomas Wang

(Rush University Medical Center, Chicago): Raj C. Shah

(University of California, San Francisco): Michelle A. Albert

(Emory University): Carlos Kase

(Centers for Disease Control and Prevention): Hubert Vesper and Julianne Botelho.

Data and Safety Monitoring Board (Voting Members): Lawrence S. Cohen, MD; Theodore Colton, ScD; Mark A. Espeland, PhD; Craig Henderson, MD; Alice H. Lichtenstein, ScD;

Rebecca A. Silliman, MD, PhD; and Nanette Wenger, MD (Chair). Ex-officio members include Josephine Boyington, PhD, MPH; Rebecca Costello, PhD; Cindy Davis, PhD; Peter Greenwald, MD; Gabriela Riscuta, MD; and Harold Seifried, PhD.

Journal Pre-proof

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Table 1. Baseline characteristics of VITAL participants, according to randomized assignment to vitamin D or placebo

Baseline characteristic	All participants	Vitamin D	
		Active	Placebo
Total number	25,871	12,927	12,944
Female sex—number (%)	13,085/25,871 (50.6)	6547 (50.6)	6538 (50.5)
Age, years, mean $\pm$ SD	67.1 $\pm$ 7.1	67.1 $\pm$ 7.0	67.1 $\pm$ 7.1
Race/ethnicity—number (%) ^a			
Non-Hispanic White	18,046/25,304 (71.3)	9013 (71.3)	9033 (71.4)
African American	5106/25,304 (20.2)	2553 (20.2)	2553 (20.2)
Hispanic (not African American)	1013/25,304 (4.0)	516 (4.1)	497 (3.9)
Asian/Pacific Islander	388/25,304 (1.5)	188 (1.5)	200 (1.6)
Native American	228/25,304 (0.9)	118 (0.9)	110 (0.9)
Other or unknown	523/25,304 (2.1)	259 (2.0)	264 (2.1)
Body mass index, kg/m ² , mean $\pm$ SD ^b	28.1 (5.7)	28.1 $\pm$ 5.7	28.1 $\pm$ 5.8
Current smoking—number (%)	1836/25,485 (7.2)	921 (7.2)	915 (7.2)
Hypertension treated with medication—number (%)	12,791/25,698 (49.8)	6352 (49.5)	6439 (50.1)
Cholesterol-lowering medication, current use—number (%)	9524/25,428 (37.5)	4822 (38.0)	4702 (36.9)
Diabetes—number (%)	3549/25,828 (13.7)	1812 (14.0)	1737 (13.4)
Current regular aspirin use—number (%) ^c	11,570/25,497 (45.4)	5756 (45.2)	5,814 (45.6)

Abbreviations: SD = standard deviation. Percentages may not sum to 100 because of rounding. There were no significant differences in the baseline characteristics between the groups.

^aRace and ethnic group were reported by participants.

^bThe body mass index is calculated as weight in kilograms divided by the square of height in meters. For body mass index, data were missing for 2.4% of the participants.

^cAt least monthly.

Adapted from Manson JE et al., *N Engl J Med* 2019; 380:33-44.

Table 2. Hazard ratios (HR) and 95% confidence intervals (CI) of primary, secondary, and other outcomes by randomized assignment to vitamin D or placebo^a

Endpoint	Vitamin D (N = 12,927)	Placebo (N = 12,944)	HR	95% CI
	<i>no. of participants w/event</i>			
Total invasive cancer^b	793	824	0.96	0.88-1.06
Cancer mortality	154	187	0.83	0.67-1.02
Breast cancer	124	122	1.02	0.79-1.31
Prostate cancer	192	219	0.88	0.72-1.07
Colorectal cancer	51	47	1.09	0.73-1.62
Cardiovascular disease (CVD), primary and secondary outcomes				
Major CVD event ^{b,c}	396	409	0.97	0.85-1.12
Expanded CVD event ^d	536	558	0.96	0.86-1.08
Total myocardial infarction	169	176	0.96	0.78-1.19
Total stroke	141	149	0.95	0.76-1.20
Cardiovascular mortality	152	138	1.11	0.88-1.40
All-cause mortality	485	493	0.99	0.87-1.12
Excluding the first two years of follow-up:				
Total invasive cancer	490	522	0.94	0.83-1.06
Cancer mortality	112	149	0.75	0.59-0.96
Major CVD event	274	296	0.93	0.79-1.09
All-cause mortality	368	384	0.96	0.84-1.11

^aAnalyses were from Cox regression models controlling for age, sex, and n-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^bPrimary outcome.

^cA composite of myocardial infarction, stroke, and cardiovascular mortality.

^dA composite of major cardiovascular events plus coronary revascularization (coronary-artery bypass grafting or percutaneous coronary intervention).

Adapted from: Manson JE et al., *N Engl J Med* 2019; 380:33-44.

Table 3. Hazard Ratios (HR) and 95% Confidence Intervals (CI) of the Primary Outcomes Comparing Vitamin D and Placebo Groups, According to Selected Prespecified Subgroups^a

Subgroup	# ppts	Total Invasive Cancer				Major Cardiovascular Events			
		Vit D	Placebo	HR (95%CI)	P, int.	Vit D	Placebo	HR (95%CI)	P, int.
		# ppts with event				# ppts with event			
Race	25,304				0.21				0.37
Non-Hispanic white	18,046	626	632	0.99 (0.89-1.11)		280	301	0.93 (0.79-1.10)	
African American	5,106	98	126	0.77 (0.59-1.01)		69	76	0.91 (0.65-1.26)	
Other	2,152	53	52	1.03 (0.70-1.51)		32	24	1.36 (0.80-2.31)	
Body Mass Index (kg/m²)	25,254				0.002				0.66
<25	7,843	206	278	0.76 (0.63-0.90)		117	115	1.07 (0.83-1.38)	
25-<30	10,122	338	323	1.04 (0.90-1.21)		152	162	0.93 (0.74-1.16)	
≥30	7,289	228	199	1.13 (0.94-1.37)		120	120	0.98 (0.76-1.26)	
Baseline serum 25(OH)D^b	15,787				0.99				0.75
<20 ng/mL	2,001	58	63	0.97 (0.68-1.39)		34	34	1.09 (0.68-1.76)	
≥20 ng/mL	13,786	459	464	0.98 (0.86-1.12)		218	216	1.00 (0.83-1.21)	
Baseline serum 25(OH)D^{b, c}	15,787				0.57				0.42
<cohort median	7,812	251	252	1.02 (0.86-1.21)		128	139	0.94 (0.74-1.20)	
≥cohort median	7,975	266	275	0.95 (0.80-1.12)		124	111	1.09 (0.84-1.41)	
Omega-3 FA randomization status^d	25,871				0.56				0.56
Placebo group	12,938	385	412	0.94 (0.82-1.08)		210	209	1.01 (0.83-1.22)	
Omega-3 FA group	12,933	408	412	0.99 (0.87-1.14)		186	200	0.93 (0.76-1.14)	

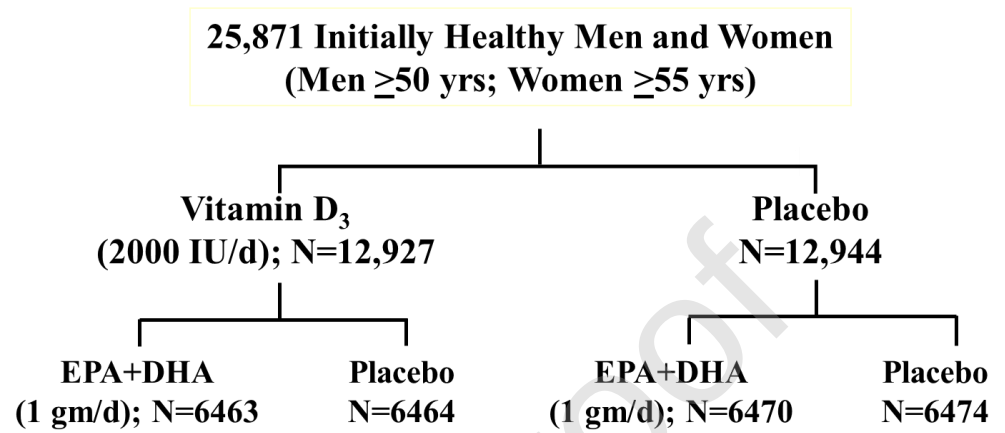
^a Analyses were from Cox regression models controlling for age, sex, and omega-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^b To convert 25(OH)D values from ng/mL to nmol/L, multiply by 2.5.

^c Median serum 25(OH)D in the cohort was 31 ng/mL.

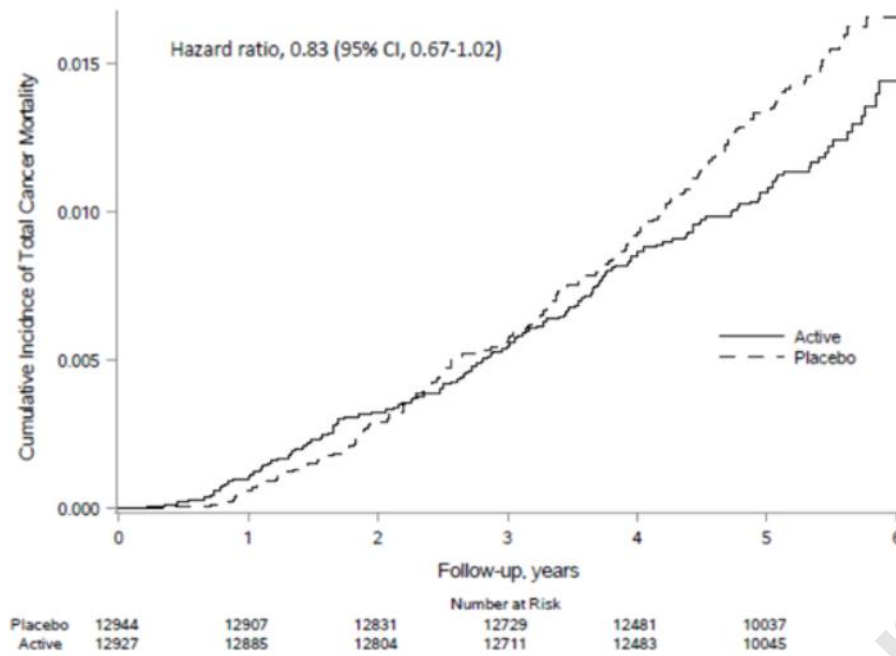
^d Randomization status in the trial.

Adapted from Manson JE et al., *N Engl J Med* 2019;380:33-44.

Figure 1. VITAL Factorial Design

Adapted from Bassuk SS et al., Contemp Clin Trials 2016; 47:235-243

Figure 2. Cumulative Incidence Rates of Cancer Mortality by Year of Follow-up: Vitamin D vs. Placebo



Source: Manson JE et al., N Engl J Med 2019;380:33-44.

Figure 3. Meta-analysis of randomized trials of vitamin D supplementation and cancer mortality (Keum et al., Ann Oncol 2019; 30(5):733-743.).

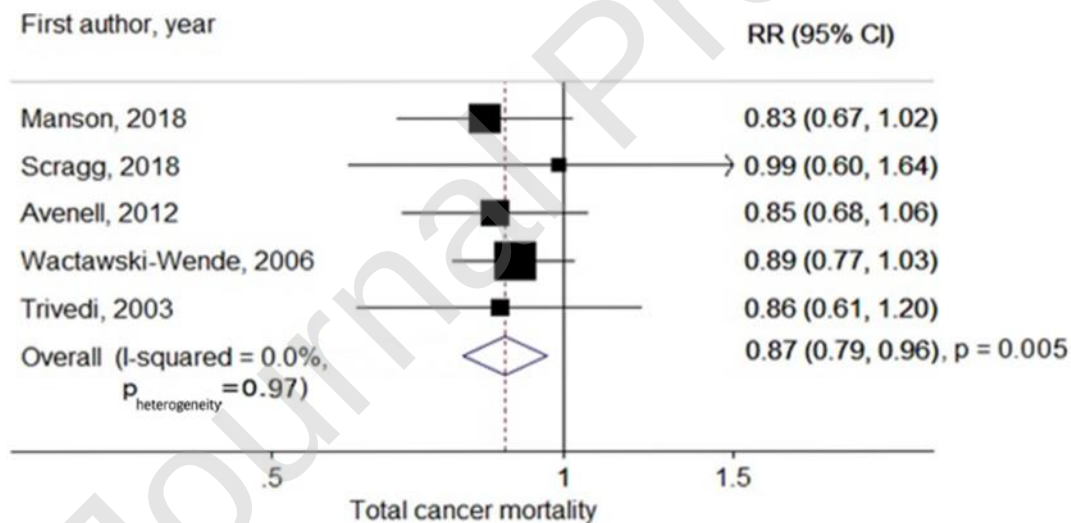
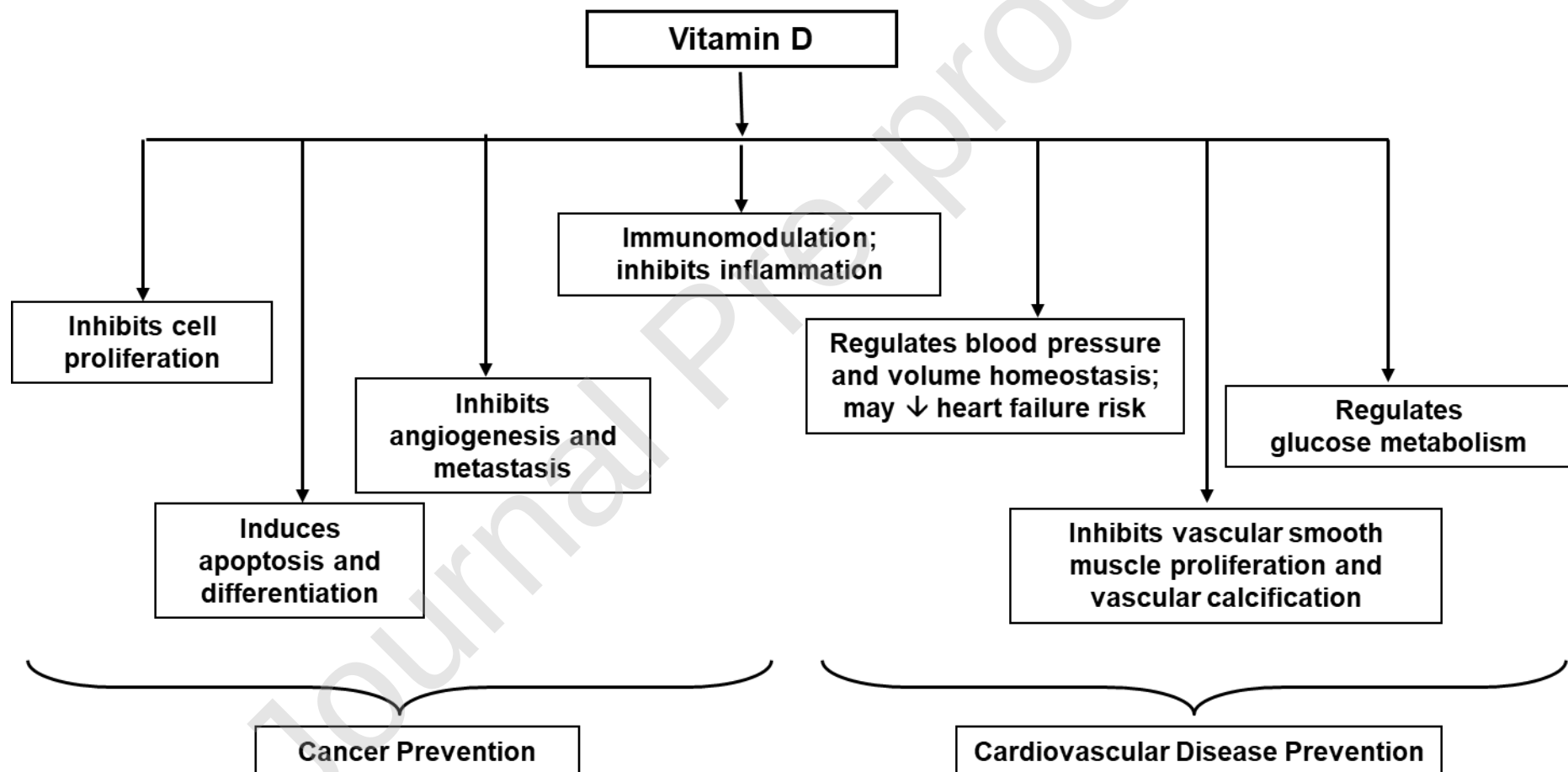


Figure 4. Mechanisms by which Vitamin D May Lower Cancer and Cardiovascular Disease Risk



Adapted from: Manson JE et al., Contemp Clin Trials 2012; 33:159-171.